

Program	Faculty	Branch/Specialization	Name of Subject	Subject Code	Marks		Credit
Ph.D	Engineering & Technology	Computer Science and Engineering	Machine learning (ML)		IM	EM	4

Unit-wise Content distribution

Unit	Contents
Unit-I	INTRODUCTION Machine learning basics: What is Machine Learning, Types and Applications of ML, Tools used, AI vs ML .Introduction to Neural Networks. Introduction to linear regression: SSE; gradient descent; closed form; normal equations; features, Introduction to classification: Classification problems; decision boundaries; nearest neighbor methods. Linear regression; SSE; gradient descent; closed form; normal equations; features Overfitting and complexity; training, validation, test data, and introduction to Matlab (II).
Unit-II	SUPERVISED LEARNING: Introduction to Supervised Learning, Supervised learning setup, LMS, Linear Methods for Classification, Linear Methods for Regression, Support Vector Machines. Basis Expansions, Model Selection Procedures Perceptron, Exponential family, Generative learning algorithms, Gaussian discriminant analysis, Naive Bayes, Support vector machines, Model selection and feature selection, Decision Tree, Ensemble methods: Bagging, boosting, Evaluating and debugging learning algorithms. Classification problems; decision boundaries; nearest neighbor methods, Probability and classification, Bayes optimal decisions Naive Bayes and Gaussian class-conditional distribution, Linear classifiers Bayes' Rule and Naive Bayes Model, Logistic regression, online gradient descent, Neural Networks Decision tree and Review for Mid-term, Ensemble methods: Bagging, random forests, boosting A more detailed discussion on Decision Tree and Boosting.
Unit-III	REINFORCEMENT LEARNING: Markov decision process (MDP), HMM, Bellman equations, Value iteration and policy iteration, Linear quadratic regulation, Linear Quadratic Gaussian, Q-learning, Value function approximation, Policy search, Reinforce, POMDPs.
Unit-IV	UNSUPERVISED LEARNING: Introduction to Unsupervised Learning: Association Rules, Cluster Analysis, Reinforcement Learning, Clustering K-means, EM. Mixture of Gaussians, Factor analysis, PCA (Principal components analysis), ICA (Independent components analysis); hierarchical agglomeration Advanced discussion on clustering and EM, Latent space methods; PCA, Text representations; naive Bayes and multinomial models; clustering and latent space models, VC-dimension, structural risk minimization; margin methods and support vector machines (SVM), Support vector machines and large-margin classifiers Time series; Markov models; autoregressive models.
Unit-V	DIMENSIONALITY REDUCTION: Feature Extraction, Singular value decomposition. Feature selection –feature ranking and subset selection, filter, wrapper and embedded methods. Machine Learning for Big data: Big Data and MapReduce, Introduction to Real World ML, Choosing an Algorithm, Design and Analysis of ML Experiments, Common Software for ML.

Textbooks/References:

1. Tom M. Mitchell, –Machine Learning, McGraw-Hill Education (India) Private Limited, 2013.
2. Ethem Alpaydin, –Introduction to Machine Learning (Adaptive Computation and Machine Learning), The MIT Press 2004.
3. Stephen Marsland, –Machine Learning: An Algorithmic Perspective, CRC Press 2009.